Electron Transport In Molecular Nanostructures

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To my parents, for they unconditional love and support,
and to Cecilia as she never let me down.

Acknowledgements

Personal

This is where you thank your advisor, colleagues, and family and friends.

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Abstract

Semiconductor materials capable of complementing Si in nanoelectronics are needed to face the recent global chip manufacturing crisis. Significant efforts have been expended in adapting graphene electronics into integrated circuits, but the various limitations inherent to its operating mechanism incentivise the study of other carbon nanostructures.

Here, I investigate molecular graphene nanostructures as a prospective nanoelectronics candidate. I began by describing the instrumentation and device designs used to collect the electrical transport data presented in this thesis. I then demonstrate that molecular graphene nanoribbons (MGNRs) can be employed as a molecular quantum dot platform at high temperatures. These findings provide the first observation of quantum events at high temperatures in a full-carbon, all-around-molecular structure at the device level.

Compared to carbon nanotubes, promoting MGNRs de-bundling when they are dispersed in organic solvent can be achieved more easily, and I describe possible strategies that could optimize the device yield and reproducibility.

I then examine the effect of adding dopant groups to the ribbon's edges. The energy shift induced by the dopant to the ribbon Fermi level is, however, weak and useful at the device level. I present the first field-effect transistor measurements of an unique class of materials in which the dopant atoms are inserted directly into the honeycomb lattice, and propose possible improvements.

Based on the research I conducted for this thesis, I believe that the future of practical MGNR-based electronics is still in its infancy. They host a multitude of physical events that should be investigated thoroughly using both STM and device-level techniques. Improving the length of the ribbon while maintaining a high degree of debundling will be crucial to the development of the first MGNR-based microprocessor.

On the basis of the research I performed in this thesis, I believe that the outlook for any practical MGNR-based electronics is distant. MGNRs host a plethora of physics phenomena which they should be first investigate thouroghly, both through STM and at the device level. The necessity of synthesising longer ribbon while mantaining a high degree of debundling will also be fundamental towards the realisation of a first MGNR-based microprocessor.

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Abbreviations

AC alternating current. 91, 100

ACE artificial concentrated evaporation. 80, 81

AFM atomic force microscope. 78

AGNR armchair graphene nanoribbon. 133

CMOS complementary metal-oxide-semiconductor. 2, 3, 10, 18, 27

CNFET carbon-nanotube field-effect transistor. 35

CNT carbon nanotube. 1, 15, 18, 25, 41, 79, 80, 108, 116, 117, 127, 133

CNTFET carbon nanotube field-effect transistor. 15, 16, 18, 23, 24

COT cotunnelling. 51

CVD chemical vapour deposition. 69, 100, 147

DAC digital-to-analog converter. 82, 101

DC direct current. 68, 82, 90, 91, 125

DFT Density Functional Theory. 130, 133, 136, 154

DIBL drain induced barrier lowering. 18

DOF degree of freedom. 55, 57

DOS density of states. 7, 8, 32, 33, 46, 127, 138, 140, 143

EBL electron-beam litography. 29, 102

EOT effective oxide thickness. 24, 42

ESD electronic discharge. 83

FET field-effect transistor. 8, 9, 15, 38, 141, 145, 149, 150, 151, 155, 170, 171, 172, 173

FWHM full width at half maximum. 115

GGA generalised-gradient approximation. 137, 159

GHB graphene Hall bar. 100, 101, 102, 103, 104, 105

GNR graphene nanoribbon. 1, 15, 27, 28, 29, 30, 33, 35, 41, 42, 43, 73, 77, 79, 80, 81, 122, 133, 136, 140, 149, 154

GNRFET graphene nanoribbon field-effect transistor. 35, 41, 42, 79

GTJ graphene tunnell junction. 67, 68, 71, 72, 73, 75, 131

hBN hexagonal boron nitride. 11, 100, 105

HOMO Highest Occupied Molecular Orbital. 30

IC integrated circuit. 2, 4

LDA local density approximation. 137, 159

LUMO Lowest Unoccupied Molecular Orbital. 30

MGNR molecular graphene nanoribbon. 107, 108, 109, 111, 113, 114, 115, 116, 117, 118, 120, 121, 122, 123, 125, 127, 128, 130, 131, 132, 134, 136, 143, 153, 155

MOSFET metal–oxide–semiconductor field-effect transistor. 19, 22

NEGF non-equilibrium Green function. 26, 138

pGNR porphyrin–GNR. 144, 145, 146, 149, 150, 152, 169

QD quantum dot. 46, 50, 51

RF radiofrequency. 12

RT room temperature. 107, 108, 109, 111

SB Schottky barrier. 16, 35, 40, 109

SET single-electron transistor. 67, 106, 107, 108, 113, 116, 122, 123, 130, 131, 132, 133, 145, 155

SS sub-threshold swing. 24

TB tight-binding. 29

TDOS tunneling density of states. 116, 118

THF tetrahydrofuran. 79

TLL Tomonaga–Luttinger liquid. 116, 117

1 Introduction

Contents

In this chapter, I will explain the rationale and context for this thesis: can we create graphene and, more broadly, graphene-derived structures that may one day offer a competitive advantage over ubiquitous Silicon in integrated circuits? To this goal, I will present a quick overview of the key all-carbon alternatives studied since the late 1990s. The chapter is organised such that it begins with a concise overview of solid-state crystal characteristics, followed by a discussion of how these properties lead to unique dispersion relations in graphene, carbon nanotubes (CNTs), and graphene nanoribbons (GNRs). Second, a selection of the most significant experimental solutions developed to date will be critically analysed.

1.1 Motivation

Silicon has been the undisputed leader in the fabrication of integrated circuits (ICs) for almost thirty years. The widespread shortage of semiconductors that began in late 2020, however, has prompted both major chipmakers and the scientific community to think more deeply and actively about which non-silicon technologies and materials are valid candidates for replacing or, more realistically, complementing silicon. Except for market leaders such as Intel, Samsung, and TSMC, long-term manufacturing costs remain largely expensive. Bloomberg reports that the fundamental costs of constructing a semiconductor manufacturing are approximately \$15 billion, with profit requirements of roughly \$3 billion to maintain market competition and avoid obsolescence [1]. These findings are also backed by a study conducted by the Boston Consulting Group last year, which ranked investments in research and development (22% of annual semiconductor sales) and capital spending (26%) as the highest in all industries [2].

The first point-contact transistor prototype developed at the beginning of the 1940s by Bardeen, Brattain, and William Shockley at Bell Labs and the creation of complementary metal-oxide-semiconductor (CMOS) techology by Atalla and Kahng, ten years later, marked the beginning of integrated components [3]. Arguably, the density scaling capabilities of this technology has been one of the main reasons responsible for the costant technological development over a time span of more than 50 years. In this scenario, Moore's law is considered not just as a dependable method for estimating the future number of transistors per integrated circuit, but also as a driver of innovation and competitiveness.

The search for new materials for integrated electronics is not just related to the race to increase the number of transistors per chip. Figure 1.1 displays several trends in microprocessor figures of merit over the past five decades. The clock frequency is the number of clock cycles (number of electrical pulses) that a CPU can execute every second and is measured in Hz. During these electrical pulses, the CPU can conduct fundamental operations such as memory access, data writing, and instruction retrieval. The clock speed (or rate) is architecture-dependent, hence it must be compared within the same architecture (x86-32/64 architecture is the benchmark architecture).

One of the most interesting positive correlation that it is possible to appreciate from the plot is, arguably, between the clock speed quantity with the power dissipated by the processor, typically static power, which is the power dissipated when a transistor is not switching. CMOS technology is well-known for low static power values for standard supply voltages (3–5) V. As the gate oxide thickness decreases, however, the probability of tunnelling for the charge carriers increases, resulting larger leakage currents and higher disspation powers [4].

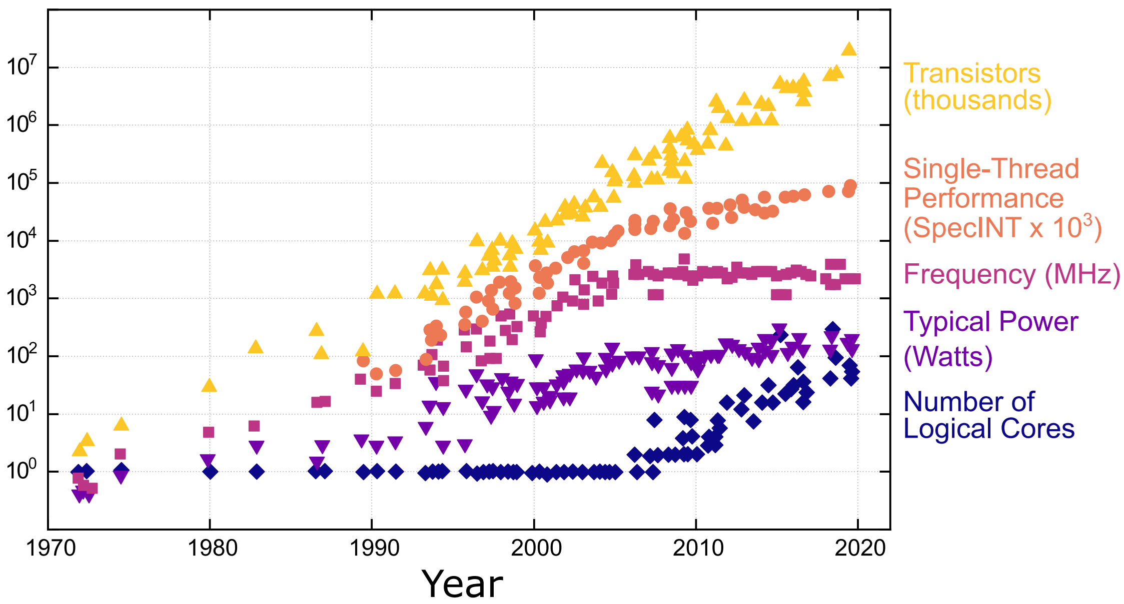

Figure 1.1: The arc of Moore's law over 42 years of microprocessor trend data. While the number of transistors have doubles about every two years, the same cannot be said for other transistors key figures. In particular, clock frequency and power dissipation capabilities have plateaued out since 2003. The released by Intel in 2005 of its first dual-core processor (Pentium D) and the ability to do Hyper-Threading sparked the number of logical cores (capability to carry out two o more processes simultaneously by a single core) per CPU. Since then, the market interest for multithreading over single-thread performance has been one of the driving factors for multiple-cores architectures in modern laptop. Data collected by K. Rupp, figure inspired by a talk given by Nvidia.

Several suggestions have envisioned graphene as one of the most robust materials against short-channel effects [5] and the possibility of integrating it into semiconductor manufacturing cycles seems within reach. Typically, these are divided into front- and back-end lines. In order to understand where this proposal comes from, the next section will outline some fundamentals properties of graphene and how they have been used in IC prototypes.

1.2 Graphene

Graphene has a hexagonal structure where the carbon atoms are hybridised sp^2. In the International Table of Crystallography, graphene is included in the symmorphic space group #191, with symmetry D_{6h} ($P6/mmm$). The triangular lattice is obtained by considering two atoms per unit cell having the respective lattice vectors:

The carbon atoms in graphene form a two-dimensional hexagonal crystal with two atoms per unit cell, as shown in Fig. 1.2:

$$\mathbf{a}_1 = \frac{a}{2}(3, \sqrt{3}),\ \mathbf{a}_2 = \frac{a}{2}(3, -\sqrt{3}), \tag{1.1}$$

with $a \simeq 1.42$ Å be the carbon-carbon bond length. Each atom belonging to one sublattice, called sublattice A, has three first neighbours belonging to the other sublattice, called sublattice B. With respect to one atom chosen from sublattice A as origin, the position of these first three neighbours in real space are given by:

$$\boldsymbol{\delta}_1 = \frac{a}{2}(1, \sqrt{3}),\ \boldsymbol{\delta}_2 = \frac{a}{2}(1, -\sqrt{3}),\ \boldsymbol{\delta}_3 = -a(1, 0), \tag{1.2}$$

While the reciprocal-lattice vectors are given may be written as:

$$\mathbf{b}_1 = \frac{2\pi}{3a}(1, \sqrt{3}),\ \mathbf{b}_2 = \frac{2\pi}{3a}(1, -\sqrt{3}), \tag{1.3}$$

In order to define the vertices of the Brillouin zone one has to apply the Bragg condition:

$$2\mathbf{k} \cdot \mathbf{G} = 0 \tag{1.4}$$

where:

$$\mathbf{G} = n\mathbf{b}_1 + m\mathbf{b}_2, \tag{1.5}$$

The first amongst these are:

$$
\begin{aligned}
&n = \pm 1, && m = 0, && \mathbf{G} = \pm \mathbf{b}_1, \\
&n = 0, && m = \pm 1, && \mathbf{G} = \pm \mathbf{b}_2, \\
&n = \pm 1, && m = \pm 1, && \mathbf{G} = \pm (\mathbf{b}_1 + \mathbf{b}_2),
\end{aligned}
$$

and all these vectors have the same modulus $|\mathbf{G}| = \frac{4\pi}{3a}$. By applying Bragg's condition in the first of the three cases we get:

$$\sqrt{3}k_x + k_y = \mp \frac{4\pi}{3a}, \tag{1.6}$$

which gets satisfied by the points:

$$K \equiv \mp \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, 1\right) \tag{1.7}$$

For the second case one gets:

$$\mp(\sqrt{3}k_x - k_y) = \frac{4\pi}{3a}, \tag{1.8}$$

which gives the points:

$$K' \equiv \pm \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, -1\right) \tag{1.9}$$

The third and last case gives the point:

$$M \equiv \pm\left(0, \pm \frac{2\pi}{3a}\right) \tag{1.10}$$

Based on the experimental evidence collected in the last 20 years, the model that seems to best describe the proprierties of graphene is the Wallace model of the band structure, which dates back in 1947, and describes graphene as a semimetal with a linear dispersion of the electronic excitations close to the Fermi energy [7]. Within the second quantization formalism ($\hbar = 1$ units), the tight-binding Hamiltonian may be written as